PROJECT PROPOSAL

Smart Transit

AI-Powered Multimodal Journey Planner

A Constraint-Aware Route Planning System for Dhaka City

An Academic Project on Graph Search,
Multimodal Routing, and Constraint-Based Optimization

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September 9, 2026

Abstract

Urban transportation in a densely populated city such as Dhaka involves multiple transportation modes, including walking, buses, and metro rail. In practical situations, travelers often have several constraints beyond simply finding a path from one location to another. For example, a user may have a limited budget, a specific arrival deadline, a preference for less walking, or a desire to minimize the number of transfers.

This project proposes **Smart Transit**, an academic prototype of an AI-powered multimodal journey planner that accepts natural-language travel requests and generates personalized route recommendations. The system will use the Google Gemini API to understand user queries and extract structured travel constraints such as origin, destination, budget, arrival deadline, and preferences. OpenStreetMap Nominatim will be used to resolve textual locations into geographical coordinates.

The core routing engine will operate on a custom multimodal transportation graph representing selected locations and connections in Dhaka. Four core algorithmic components will be combined for different routing purposes: **Dijkstra's Algorithm** for reliable shortest-path search, **A* Search** for heuristic-guided route search, **Yen's K-Shortest Paths** for generating alternative route candidates, and **Multi-Criteria Route Optimization** for ranking feasible routes according to time, fare, walking distance, and transfers. A constraint and ranking engine will then filter infeasible routes and identify the *Best Match*, *Fastest*, and *Cheapest* options.

The primary objective of this project is not to build a production-grade navigation platform, but to demonstrate how multiple classical search and optimization algorithms can be combined with modern APIs to solve a practical multimodal route-planning problem.

1. Introduction

Finding a route between two locations is a classical graph-search problem. Traditional shortest-path algorithms generally optimize a single objective such as distance, travel time, or cost. However, real-world transportation decisions are usually multi-objective.

For example, consider the following request:

> "I want to travel from UIU to Dhaka University within 100 BDT, arrive before 9 AM, and avoid excessive walking."

A route that is fastest may exceed the user's budget. A route that is cheapest may require more walking or take longer. Similarly, a route with fewer transfers may not be the fastest or cheapest.

Smart Transit addresses this problem by combining natural-language query understanding with graph-based route search and constraint-aware ranking.

The system is designed as an academic prototype focused on demonstrating algorithmic concepts. Rather than attempting to model the complete transportation infrastructure of Dhaka, the project will use a selected set of important locations, transit stops, and transportation connections to construct a manageable multimodal graph.

2. Problem Statement

Most basic route-planning systems focus on finding one optimal path according to a predefined metric. Such an approach is insufficient when a traveler provides multiple constraints and preferences.

The problem addressed by this project can be formulated as follows:

> Given a natural-language journey request containing an origin, destination, budget, arrival deadline, and optional travel preferences, generate multiple feasible multimodal routes and select the routes that best satisfy the user's requirements.

The system must answer questions such as:

- Which route is the fastest?
- Which route is the cheapest?
- Which route requires the fewest transfers?
- Which feasible route best matches the user's preferences?
- Why was a particular route selected?

3. Motivation

Dhaka's transportation environment makes multimodal journey planning particularly relevant. A traveler may need to combine walking, buses, and metro rail in a single journey.

Users also express transportation requirements naturally rather than as formal parameters. Instead of entering separate values into a form, a user may simply write:

> "I have 80 taka and need to reach Farmgate before 8:30. I don't want to walk much."

Natural-language processing can transform such input into structured constraints, while graph algorithms can solve the actual routing problem.

Therefore, the project provides an opportunity to integrate:

1. Natural-language understanding,
2. Geographical location resolution,
3. Graph representation,
4. Classical search algorithms,
5. Constraint handling, and
6. Multi-criteria route ranking.

4. Objectives

The main objectives of Smart Transit are:

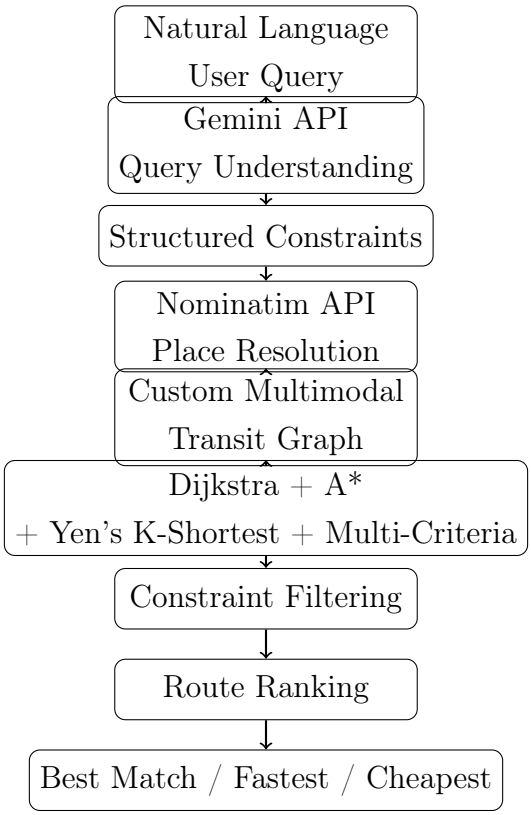
1. To develop a natural-language interface for journey planning.
2. To extract travel constraints such as origin, destination, budget, deadline, and preferences from user queries.
3. To resolve textual place names into geographical coordinates.
4. To construct a simplified multimodal transportation graph for selected areas of Dhaka.
5. To implement and integrate four complementary graph-search and optimization components.
6. To use A* Search for heuristic-guided route exploration.
7. To use Dijkstra's Algorithm as a reliable shortest-path baseline.
8. To use Yen's K-Shortest Paths algorithm to generate multiple alternative route candidates.
9. To use Multi-Criteria Route Optimization to rank feasible routes according to user preferences.
10. To develop a constraint engine that filters routes based on budget and arrival deadline.
11. To rank feasible routes and present Best Match, Fastest, and Cheapest recommendations.
12. To provide a simple explanation of why a particular route was recommended.

5. Proposed System

Smart Transit will consist of several interconnected components.

5.1 High-Level Workflow

The complete workflow is:



6. System Components

6.1 6.1 Natural Language Query Processing

The system will accept user requests in either English or Bangla.

Example:

"UIU DU , 100 9 ."

The Google Gemini API will transform the natural-language request into a structured representation.

Example:

```
{
    "origin": "UIU",
    "destination": "Dhaka University",
```

```
    "budget": 100,
    "arrival_deadline": "09:00",
    "preferences": {
        "walking": "low",
        "transfers": "medium"
    }
}
```

The Gemini API will only perform query understanding. It will not be responsible for calculating the optimal route.

6.2 6.2 Place Resolution

The OpenStreetMap Nominatim API will be used to resolve textual locations into latitude and longitude coordinates.

For example:

```
"United International University"
       ->
Latitude, Longitude
```

The coordinates will then be mapped to the nearest relevant node in the custom transit graph.

6.3 6.3 Multimodal Transit Graph

The routing environment will be represented as a weighted graph.

Each node may represent:

- A landmark,
- A bus stop,
- A metro station, or
- A transfer point.

Each edge may represent:

- Walking,
- Bus travel,
- Metro travel, or
- A transfer between modes.

Each edge will contain simplified attributes such as:

```
Source
Destination
Mode
Distance
Estimated Time
Fare
```

Example:

```
{
    "source": "UIU",
    "destination": "Kuril",
    "mode": "walk",
    "time": 8,
    "distance": 0.6,
    "fare": 0
}
```

The graph will be intentionally limited to selected locations in Dhaka to keep the academic implementation manageable.

7. Algorithm Design

The central routing engine will use four complementary algorithmic components. Each component has a specific responsibility in generating or selecting multimodal journey alternatives.

The four core components are:

1. Dijkstra's Algorithm
2. A* Search
3. Yen's K-Shortest Paths
4. Multi-Criteria Route Optimization

7.1 7.1 Dijkstra's Algorithm

Dijkstra's Algorithm will provide a reliable shortest-path baseline on the weighted multimodal graph. For non-negative edge weights, it finds the path that minimizes the total accumulated cost.

$$w(e) \geq 0$$

The edge weight may represent travel time, generalized travel cost, or another non-negative routing objective.

Dijkstra → Reliable Shortest-Path Baseline

7.2 7.2 A* Search

A* Search will provide heuristic-guided route exploration between the origin and destination.

$$f(n) = g(n) + h(n)$$

where $g(n)$ is the cost from the origin to node n and $h(n)$ is the estimated remaining cost to the destination. For geographically represented nodes, straight-line geographic distance can be used as a heuristic when it is consistent with the selected edge-cost formulation.

A* → Heuristic-Guided Route Search

7.3 7.3 Yen's K-Shortest Paths

A single shortest route is often insufficient for a journey planner. Yen's K-Shortest Paths algorithm will therefore generate multiple loopless route alternatives.

For a requested value of K, the algorithm produces a ranked set:

$$R_1, R_2, \ldots, R_K$$

where R_1 is the shortest route and subsequent routes provide alternative path choices.

This candidate pool is important because a passenger may prefer a slightly longer route if it is cheaper, requires less walking, or uses fewer transfers.

Yen's K-Shortest Paths → Alternative Route Generation

7.4 7.4 Multi-Criteria Route Optimization

The final recommendation should not be selected using distance alone. Candidate routes will be evaluated using travel time, fare, walking distance, and number of transfers.

Because these attributes have different units, the values will first be normalized. A weighted score can then be calculated:

$$Score(R) = w_t T(R) + w_f F(R) + w_w W(R) + w_x X(R)$$

where T, F, W, and X represent normalized time, fare, walking, and transfer measures.

The route with the lowest score among feasible candidates will be selected as the Best Match.

Multi-Criteria Optimization → Preference-Based Route Ranking

7.5 7.5 Pareto Analysis as an Evaluation Layer

Pareto analysis will be used as a supporting multi-objective evaluation technique rather than as one of the four core routing algorithms.

A route is Pareto-dominated if another feasible route is no worse on all selected criteria and strictly better on at least one criterion. The non-dominated routes form a Pareto set. This helps explain trade-offs between travel time, fare, walking distance, and transfers.

8. Why the Four Algorithms Are Combined

The four core components are designed as a pipeline rather than as four unrelated demonstrations.

Table 1: Role of Algorithms in Smart Transit

Component	**Primary Role**	**Reason for Use**
Dijkstra	Shortest-path baseline	Provides a reliable optimal path under non-negative edge costs.
A*	Heuristic-guided search	Uses destination information to focus route exploration.
Yen's K-Shortest Paths	Alternative generation	Produces multiple candidate routes for passenger choice and ranking.
Multi-Criteria Optimization	Route ranking	Balances time, fare, walking, and transfers according to user preferences.

Dijkstra + A* + Yen's K-Shortest + Multi-Criteria

Pareto analysis and constraint filtering are supporting evaluation and decision layers, not additional core routing algorithms.

9. Algorithm Blending Pipeline

The routing pipeline separates candidate generation from route selection.

1. The user provides a natural-language journey request.
2. Gemini extracts the origin, destination, budget, deadline, and travel preferences.
3. Nominatim resolves the textual locations into geographical coordinates.
4. The coordinates are mapped to nodes in the multimodal transit graph.
5. Dijkstra generates a reliable shortest-path baseline.
6. A* generates a heuristic-guided route candidate.
7. Yen's K-Shortest Paths generates additional alternative routes.
8. All generated routes are collected into a candidate route pool.
9. The Constraint Engine removes routes that violate hard constraints such as budget and arrival deadline.
10. Multi-Criteria Optimization normalizes and scores the remaining routes using time, fare, walking distance, and transfers.
11. Pareto analysis can identify non-dominated alternatives and explain trade-offs.
12. Finally, the system presents Best Match, Fastest, and Cheapest recommendations.

$$\text{Transit Graph} \rightarrow \begin{cases} \text{Dijkstra} & \rightarrow \text{Shortest-Path Baseline} \\ A^* & \rightarrow \text{Heuristic-Guided Route} \\ \text{Yen's K-Shortest} & \rightarrow \text{Alternative Routes} \end{cases}$$

$$\text{Candidate Routes} \rightarrow \text{Constraint Filtering} \rightarrow \text{Multi-Criteria Ranking} \rightarrow \text{Pareto Analysis}$$

10. Constraint Engine

After candidate generation, the system will apply hard constraints provided by the user. For example:

$$Fare(route) \leq Budget$$

and

$$ArrivalTime(route) \leq Deadline$$

If a route violates the budget:

```
if route.fare > user_budget:
    reject(route)
```

If a route arrives after the deadline:

```
if route.arrival_time > deadline:
    reject(route)
```

This ensures that a route that does not satisfy a user's mandatory requirements is not selected as the Best Match.

11. Route Ranking

After constraint filtering, the remaining routes will be ranked using three primary categories.

11.1 11.1 Fastest Route

The fastest route is:

$$R_{fastest} = \arg\min_{R} TravelTime(R)$$

This route minimizes total estimated travel time.

11.2 11.2 Cheapest Route

The cheapest route is:

$$R_{cheapest} = \arg\min_{R} Fare(R)$$

This route minimizes the total estimated fare.

11.3 11.3 Best Match Route

The Best Match route will be selected according to user preferences.

A simplified weighted objective can be:

$$Score(R) = w_t T(R) + w_f F(R) + w_w W(R) + w_x X(R)$$

The route with the lowest score among feasible routes will be selected.

For example, if the user strongly prefers less walking, the walking weight w_w will be increased.

Thus:

$$\text{User Preference} \rightarrow \text{Weights} \rightarrow \text{Route Score} \rightarrow \text{Best Match}$$

12. Example Scenario

Consider the following user query:

> "I want to go from UIU to Dhaka University. My budget is 100 BDT, I need to arrive before 9 AM, and I want to minimize walking."

Gemini may extract:

```
Origin: UIU
Destination: Dhaka University
Budget: 100 BDT
Deadline: 09:00 AM
Walking Preference: Low
```

The algorithms may generate:

Table 2: Example Candidate Routes

Route	**Time**	**Fare**	**Walking**	**Transfers**
A	45 min	90 BDT	700 m	2
B	61 min	55 BDT	800 m	1
C	52 min	75 BDT	400 m	1

All three routes satisfy the budget and deadline constraints.

Therefore:

- Route A can be selected as **Fastest**.
- Route B can be selected as **Cheapest**.
- Route C can be selected as **Best Match** because it provides a strong balance between time, cost, walking distance, and transfers.

13. API Integration

13.1 13.1 Google Gemini API

Gemini will be used exclusively for natural-language query understanding.

Its responsibilities are:

- Intent understanding,
- Origin extraction,
- Destination extraction,
- Budget extraction,
- Deadline extraction,
- Preference extraction.

The routing algorithms will remain independent of Gemini.

13.2 13.2 OpenStreetMap Nominatim API

Nominatim will be used for geocoding and place resolution.

Its responsibilities are:

- Converting place names into coordinates,
- Resolving common place names,
- Supporting the mapping of user locations to graph nodes.

14. Technology Stack

Table 3: Technology Stack

Component	**Technology**
Programming Language	Python
Natural Language Processing	Google Gemini API
Geocoding	OpenStreetMap Nominatim API
Graph Representation	Python / NetworkX or Custom Graph
Routing Algorithms	Dijkstra, A*, Yen's K-Shortest Paths
Optimization	Multi-Criteria Route Ranking
Backend	Flask or FastAPI
Frontend	HTML, CSS, JavaScript
Map Visualization	Leaflet.js / OpenStreetMap
Data Storage	JSON or SQLite

15. Data Design

A simplified transportation dataset will be created for selected locations in Dhaka.

Example locations may include:

- United International University,
- Bashundhara,
- Kuril,
- Banani,
- Mohakhali,
- Farmgate,
- Agargaon,
- Shahbag,
- Dhaka University,
- Motijheel.

The exact set of nodes may be adjusted during implementation.

Each connection will contain:

```
{
    "source": "...",
    "destination": "...",
    "mode": "...",
    "distance": "...",
    "time": "...",
    "fare": "..."
}
```

The data will be kept intentionally small and manageable for the academic prototype.

16. Functional Requirements

The system shall:

1. Accept natural-language travel queries.
2. Support both English and Bangla queries where possible.
3. Extract origin and destination.
4. Extract budget constraints.
5. Extract arrival deadlines.
6. Extract basic user preferences.
7. Resolve locations using Nominatim.
8. Construct or load the transit graph.
9. Execute A* Search.
10. Execute Dijkstra's Algorithm.
11. Generate alternative routes using Yen's K-Shortest Paths.
12. Rank candidate routes using Multi-Criteria Optimization.
13. Filter routes using hard constraints.
14. Rank feasible routes.
15. Display Fastest, Cheapest, and Best Match routes.
16. Provide a basic explanation of the recommendation.

17. Non-Functional Requirements

17.1 Usability

The interface should be simple enough for users to provide travel requirements using natural language.

17.2 Performance

The selected graph will be small enough that route calculations can be completed quickly on a standard computer.

17.3 Maintainability

The system will separate API processing, graph construction, algorithms, constraints, and ranking into independent modules.

17.4 Interpretability

The system should provide understandable reasons for route recommendations rather than presenting only a route identifier.

18. System Limitations

The proposed system is an academic prototype and therefore has several limitations.

- The transportation graph will cover only selected locations rather than the complete Dhaka transportation network.
- Bus routes and estimated travel times may be based on static or manually prepared data.
- The system will not provide production-level real-time navigation.
- Real-time traffic prediction is outside the core scope.
- Weather integration, if implemented, will be treated as an optional enhancement.
- Nominatim may return ambiguous results for certain place names.
- The system will provide estimated route information rather than guaranteed real-world travel times.

These limitations are intentional to keep the project feasible within an academic semester.

19. Evaluation Plan

The system will be evaluated using a set of predefined travel scenarios.

The primary evaluation metrics will include:

1. Total estimated travel time,
2. Total fare,
3. Walking distance,
4. Number of transfers,
5. Algorithm execution time,
6. Constraint satisfaction rate.

Example test cases include:

Table 4: Planned Test Cases

Case	Description
1	Basic origin-to-destination query.
2	Query with a maximum budget.
3	Query with an arrival deadline.
4	Query with a low-walking preference.
5	Query with multiple simultaneous constraints.
6	English natural-language query.
7	Bangla natural-language query.

The output of the algorithms will be compared to verify whether each algorithm fulfills its intended role.

20. Expected Results

The expected outcome is a working academic prototype capable of transforming a natural-language journey request into personalized route recommendations.

For a query such as:

"UIU to DU, under 100 BDT, arrive before 9 AM, less walking."

the system is expected to produce:

```
BEST MATCH
Route: Walk -> Bus -> Metro -> Walk
Estimated Time: 52 minutes
Estimated Fare: 75 BDT
Walking: 400 meters
Transfers: 1

FASTEST
Estimated Time: 45 minutes

CHEAPEST
Estimated Fare: 55 BDT
```

The system should also explain why the Best Match route satisfies the user's requirements.

21. Project Novelty

The novelty of the project does not lie in developing a new graph-search algorithm. Instead, the project demonstrates the practical combination of established algorithms for different aspects of a personalized transportation problem.

The main contribution is the integration of:

Natural Language Understanding + Geocoding + Multimodal Graph + Multiple Search Algorithms + Constraint-Based Ranking

In particular, the system combines:

$$Dijkstra \rightarrow Reliable\ Shortest\text{-}Path\ Baseline$$

$$A^* \rightarrow Heuristic\text{-}Guided\ Search$$

$$Yen's\ K\text{-}Shortest \rightarrow Alternative\ Route\ Generation$$

$$Multi\text{-}Criteria \rightarrow Preference\text{-}Based\ Ranking]$$

This combination allows the system to provide multiple meaningful route alternatives instead of returning only a single shortest path.

22. Project Scope

The project will focus on the following core features:

1. Natural-language travel query processing.
2. Selected multimodal transportation network of Dhaka.
3. Walking, bus, and metro modes.
4. Four routing/search algorithms.
5. Budget and deadline constraints.
6. Walking and transfer preferences.
7. Route ranking.
8. Basic route explanation.

The project will not attempt to reproduce the full functionality of commercial navigation platforms.

23. Future Work

The prototype can be extended in future work through:

- Real-time traffic integration,
- Real-time bus tracking,
- GTFS-based public transportation schedules,
- Dynamic route re-planning,
- Machine-learning-based travel-time prediction,
- User-history-based personalization,
- Larger-scale Dhaka transportation graphs,
- Real-time transportation disruption detection.

These features are outside the core academic scope of the current project.

24. Project Timeline

Sequence	Task
1	Requirement analysis and project planning
2	Literature review and graph design
3	Transportation data collection
4	Gemini API integration
5	Nominatim integration and location mapping
6	Dijkstra and BFS implementation
7	A* implementation
8	Local Search implementation
9	Constraint and ranking engine
10	Backend and frontend integration
11	Testing and algorithm evaluation
12	Final documentation, presentation, and demonstration

25. Conclusion

Smart Transit proposes an academic prototype for personalized multimodal journey planning in Dhaka. The system combines modern API-based natural-language and location services with classical graph-search and optimization algorithms.

The Google Gemini API will be responsible for understanding natural-language travel requirements, while OpenStreetMap Nominatim will resolve user-provided locations. The actual route computation will be performed by a custom graph-based routing engine.

The central algorithmic contribution is the combination of four complementary routing and optimization components with clearly separated responsibilities. Dijkstra's Algorithm will provide a reliable shortest-path baseline, A* Search will perform heuristic-guided route search, Yen's K-Shortest Paths will generate multiple route alternatives, and Multi-Criteria Optimization will rank feasible routes according to time, fare, walking distance, and transfers. A constraint engine will ensure that recommendations satisfy mandatory requirements such as budget and arrival deadline.

The final system will provide three useful perspectives on a journey: **Fastest**, **Cheapest**, and **Best Match**. Therefore, instead of treating route planning as a simple shortest-path problem, Smart Transit demonstrates how multiple algorithms can be combined to support a more practical and personalized transportation decision.